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NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2026

Even Semester - 2025-26

IT202 - OBJECT ORIENTED PROGRAMMING WITH JAVA

Time: 3 Hrs

Maximum Marks : 60

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Part A

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1. What is the purpose of the 'final' keyword when applied to a class? (1) C02 BL2 PO3
(a) It cannot be instantiated (b) It cannot be inherited
(c) It cannot have methods (d) It is abstract
2. What is the output of: System.out.println(10/3) in Java? (1) C02 BL2 PO3
(a) 3.33 (b) 3
(c) 3.0 (d) Compile error
3. Which method returns the number of characters in a String? (1) C02 BL1 PO3
(a) size() (b) count()
(c) length() (d) capacity()
4. Which class is primarily used for reading data from a file in Java? (1) C02 BL1 PO3
(a) FileWriter (b) FileReader
(c) FileStream (d) FileInputStream
5. Which of the following is NOT a feature of OOP? (1) C01 BL1 PO1
(a) Inheritance (b) Polymorphism
(c) Compilation (d) Encapsulation
6. Which of the following is a checked exception? (1) C03 BL2 PO2
(a) NullPointerException (b) ArrayIndexOutOfBoundsException
(c) IOException (d) ArithmeticException

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|--|-------------------------|-----|-----|-----|
| 7. Which interface must be implemented to create a thread in Java? | (1) | C02 | BL1 | PO1 |
| (a) Runnable | (b) Threadable | | | |
| (c) Serializable | (d) Callable | | | |
| 8. What does JDK stand for? | (1) | C01 | BL1 | PO1 |
| (a) Java Development Kit | (b) Java Deployment Kit | | | |
| (c) Java Design Kit | (d) Java Debug Kit | | | |
| 9. Which method must be overridden when implementing the Runnable interface? | (1) | C02 | BL1 | PO3 |
| (a) start() | (b) execute() | | | |
| (c) run() | (d) init() | | | |
| 10. What is the default value of an int variable in Java? | (1) | C01 | BL1 | PO1 |
| (a) null | (b) undefined | | | |
| (c) 0 | (d) 1 | | | |
| 11. Which is NOT a valid access specifier in Java? | (1) | C01 | BL1 | PO1 |
| (a) public | (b) private | | | |
| (c) protected | (d) friend | | | |
| 12. Which keyword is used to inherit a class in Java? | (1) | C02 | BL1 | PO3 |
| (a) implements | (b) extends | | | |
| (c) inherits | (d) super | | | |

Part B**(Answer any four of the following) 5x4=20**

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| 13. Explain encapsulation and data hiding in Java with a suitable example. | (5) | C01 | BL2 | PO1 |
| 14. Differentiate between Abstract Class and Interface in Java. When would you use each? | (5) | C03 | BL4 | PO2 |
| 15. What is compile-time and runtime polymorphism? Give one example each. | (5) | C03 | BL4 | PO2 |
| 16. Explain exception handling in Java. Illustrate with a program using try, catch, and finally. | (5) | C03 | BL3 | PO4 |
| 17. Explain String and StringBuffer classes in Java with their key methods and differences. | (5) | C02 | BL6 | PO3 |

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| 18. Explain constructor overloading in Java with a complete program. | (5) | CO2 | BL3 | PO3 |
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Part C**(Answer any three of the following) 3x10=30**

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| 19. Explain the fundamental concepts of OOP and provide Java examples for each concept. | | | | |
| (a) class | (2) | CO1 | BL2 | PO1 |
| (b) object | (2) | CO1 | BL2 | PO1 |
| (c) inheritance | (2) | CO1 | BL2 | PO1 |
| (d) polymorphism | (2) | CO1 | BL2 | PO1 |
| (e) encapsulation | (2) | CO1 | BL2 | PO1 |
| 20. (a) Distinguish between local variable, instance variable, and class (static) variable in Java. Also explain the 'this' keyword. | (4) | CO4 | BL3 | PO3 |
| (b) Write a complete Java program with a BankAccount class demonstrating local variable, instance variable, static variable, constructor with 'this' keyword. | (6) | CO2 | BL3 | PO3 |
| 21. (a) Explain user-defined exception classes in Java. | (2) | CO4 | BL3 | PO4 |
| (b) Write a complete program demonstrating: (i) custom exception creation, (ii) use of throw and throws, (iii) exception with arguments, and (iv) use of finally. | (6) | CO4 | BL3 | PO4 |
| (c) Also explain the exception class hierarchy. | (2) | CO4 | BL3 | PO4 |
| 22. (a) What is Serialization? | (3) | CO5 | BL6 | PO1 |
| (b) Write a Java program to write an object to a file and read it back. | (7) | CO5 | BL6 | PO1 |
| 23. (a) Explain multithreading in Java. | (2) | CO3 | BL3 | PO4 |
| (b) Write a program to create multiple threads using both Thread class and Runnable interface. | (5) | CO3 | BL3 | PO2 |
| (c) Explain the thread life cycle with all states. | (3) | CO3 | BL3 | PO4 |